HAC24

24-bit, 192 kHz stereo audio codec with headphone driver

Features

- ADC signal-to-noise ratio: 104 dB (A-weighted)
- DAC signal-to-noise ratio: 110 dB (A-weighted)
- THD+N: –95 dB
- Sample rates from 8 kHz to 192 kHz
- Digital audio interfaces: I²S, left-justified and TDM
- Control interface: I²C, address 0x1A (0x1B when CS pin is high)
- Analog supply AVDD: 3.0 V to 3.6 V (3.3 V nominal); digital core DVDD: 1.65 V to 1.95 V
- Headphone driver: 60 mW into 32 Ω at 3.3 V
- Playback power consumption: 18 mW
- QFN-32 (5×5 mm) package

General description

The HAC24 is a high-performance stereo codec for portable audio. It includes a programmable gain amplifier, a headphone amplifier and a low-latency digital filter option.

Applications

Portable audio, Voice assistants, Conferencing, Musical instruments.

Ordering information

Order code	Package	Packing	Temperature grade
HAC24-QFN32-T	QFN-32 (5×5 mm)	Tape & reel, 3000	–40 °C to 85 °C

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

Parameter	Min	Max	Unit
AVDD to AGND	–0.3	4.0	V
DVDD to DGND	–0.3	2.2	V
Storage temperature	–65	150	°C

Recommended operating conditions

Parameter	Min	Typ	Max	Unit
Analog supply AVDD	3.0	3.3	3.6	V
Digital core DVDD	1.65	1.8	1.95	V
Operating temperature	–40	—	85	°C

Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

Parameter	Conditions	Min	Typ	Max	Unit
ADC SNR	A-weighted, 1 kHz	—	104	—	dB
DAC SNR	A-weighted, 1 kHz	—	110	—	dB
THD+N	–1 dBFS	—	–95	—	dB
Headphone output power	32 Ω load, AVDD = 3.3 V	—	60	—	mW
Maximum sample rate		—	192	—	kHz
I²C address (CS = GND)	7-bit		0x1A		
I²C address (CS = DVDD)	7-bit		0x1B		

Pin configuration and functions

Pin	Name	Type	Description
1	MCLK	Input	Master clock.
2	BCLK	I/O	Bit clock.
3	LRCLK	I/O	Left/right (frame) clock.
4	SDIN	Input	Serial data in (DAC).
5	SDOUT	Output	Serial data out (ADC).
12	SCL	Input	I²C clock.
13	SDA	I/O	I²C data.
14	CS	Input	I²C address select.

Package and thermal information

Package	Thermal resistance θJA	Dimensions
QFN-32 (5×5 mm)	θJA = 34 °C/W	5.0 × 5.0 mm

Application information

Provide a low-noise AVDD (for example from the HM3300 regulator) and keep analog and digital grounds joined at a single point under the device.

Revision history

Revision	Date	Change
Rev A	2024-04-22	Initial release
Rev B	2025-02-03	Added TDM mode